From the many-body wave function to densities

First and second quantization; Hartree and exchange energies

Supplement to FYS4480/9480, Week 42, slides 31–34

22 October 2026

Detailed derivations for fixed-particle-number fermion states.

The question: what information determines the energy?

For a normalized antisymmetric N-fermion wave function, consider

$$\widehat{H} = \sum_{i=1}^{N} \widehat{h}(i) + \sum_{i<j} v(\mathbf{r}_i, \mathbf{r}_j), \qquad \widehat{h} = -\frac{\hbar^2}{2m}\nabla^2 + v_{\text{ext}}(\mathbf{r}).$$

We assume a symmetric, **local, spin-independent** two-body interaction.

1. Derive the one- and two-body reduced density matrices (RDMs).
2. Recover the same objects with field and creation/annihilation operators.
3. Express $E = \langle \widehat{H} \rangle$ exactly using these RDMs.
4. For one determinant, derive the Hartree and exchange contributions.

The distinction to keep throughout

The local density $n(\mathbf{r})$, the one-body RDM γ, and the two-body RDM Γ contain different amounts of information.

Coordinates, normalization, and argument order

$$x = (\mathbf{r}, \sigma), \qquad \int \mathrm{d}x = \sum_{\sigma} \int \mathrm{d}^3 r, \qquad \int \mathrm{d}x_1 \cdots \mathrm{d}x_N \, |\Psi(x_1, \ldots, x_N)|^2 = 1.$$

Antisymmetry means $\Psi(\ldots, x_i, \ldots, x_j, \ldots) = -\Psi(\ldots, x_j, \ldots, x_i, \ldots)$.

Convention used in this supplement: ket coordinates first

$$\gamma(x; x') = N \int \mathrm{d}X \, \Psi(x, X) \Psi^*(x', X), \qquad X = (x_2, \ldots, x_N).$$

The unprimed coordinate belongs to Ψ; the primed coordinate to Ψ^*.

$$n(x) = \gamma(x; x), \qquad n(\mathbf{r}) = \sum_{\sigma} n(\mathbf{r}\sigma).$$

The spin-orbital density $n(x)$ and the spin-summed spatial density $n(\mathbf{r})$ must not be confused. The week-42 notation on slide 31 is $\rho^{(1)}(x'; x) = \gamma(x; x')$.

A necessary convention check: counting pairs

Define the **ordered-pair** RDM with prefactor $N(N-1)$:

$$\Gamma(1,2;1',2') = N(N-1)\int \mathrm{d}X\, \Psi(1,2,X)\Psi^*(1',2',X).$$

Here 1 abbreviates x_1, and $X = (x_3,\ldots,x_N)$.

Convention	Trace	Interaction energy, local v
Ordered Γ	$N(N-1)$	$\frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\, v\,\Gamma(x,y;x,y)$
Distinct-pair $\rho^{(2)} = \Gamma/2$	$N(N-1)/2$	$\int \mathrm{d}x\,\mathrm{d}y\, v\,\rho^{(2)}(x,y;x,y)$

Reconciling the supplied slides

Slide 32 defines the second convention; its field-operator expression and slide 33's factorization use the first. We use Γ consistently and keep the factor $1/2$ in E_2. With $\rho^{(2)} = \Gamma/2$, every two-body factorization below must also be divided by two.

First quantization I: derive the local one-body density

The number-density operator is

$$\widehat{n}(x) = \sum_{i=1}^{N} \delta(x - x_i), \qquad \delta(x - x_i) = \delta^{(3)}(\mathbf{r} - \mathbf{r}_i)\delta_{\sigma\sigma_i}.$$

Taking its expectation value gives

$$\begin{aligned} n(x) &= \sum_i \int \mathrm{d}x_1 \cdots \mathrm{d}x_N \, |\Psi(x_1, \ldots, x_N)|^2 \delta(x - x_i) \\ &= N \int \mathrm{d}x_2 \cdots \mathrm{d}x_N \, |\Psi(x, x_2, \ldots, x_N)|^2. \end{aligned}$$

All N terms agree because $|\Psi|^2$ is invariant under relabeling particles.

$$\int \mathrm{d}x \, n(x) = N, \qquad \langle \widehat{N}_\Omega \rangle = \int_\Omega \mathrm{d}^3 r \, n(\mathbf{r}).$$

n/N is a normalized one-particle probability density; n itself is a **particle-number density**.

First quantization II: the off-diagonal one-body RDM

For a one-body kernel $a(x';x) = \langle x'|\hat{a}|x\rangle$,

$$\langle \textstyle\sum_i \hat{a}(i)\rangle = N \int \mathrm{d}X\, \mathrm{d}x'\, \mathrm{d}x\, \Psi^*(x',X) a(x';x) \Psi(x,X)$$
$$= \int \mathrm{d}x'\, \mathrm{d}x\, a(x';x)\gamma(x;x').$$

Thus the natural definition is

$$\boxed{\gamma(x;x') = N \int \mathrm{d}X\, \Psi(x,X)\Psi^*(x',X).}$$

Hermiticity and trace

$$\gamma(x;x')^* = \gamma(x';x), \quad \int \mathrm{d}x\, \gamma(x;x) = N.$$

Natural spin orbitals

$$\int \mathrm{d}x' \gamma(x;x') u_k(x') = n_k u_k(x),$$
$$0 \leq n_k \leq 1, \qquad \sum_k n_k = N.$$

The exact one-body energy: kinetic and external terms

For a local one-body Hamiltonian, applying the previous kernel identity gives

$$E_1 = T[\gamma] + E_{\mathrm{ext}}[n],$$

$$\boxed{T[\gamma] = -\frac{\hbar^2}{2m}\int \mathrm{d}x\ \left[\nabla_{\mathbf{r}}^2\gamma(x;x')\right]_{x'=x},} \qquad E_{\mathrm{ext}}[n] = \int \mathrm{d}^3 r\, v_{\mathrm{ext}}(\mathbf{r})n(\mathbf{r}).$$

The derivative acts on the ket coordinate *before* the diagonal is taken. With vanishing boundary terms, integration by parts also gives

$$T[\gamma] = \frac{\hbar^2}{2m}\int \mathrm{d}x\ \left[\nabla_{\mathbf{r}}\cdot\nabla_{\mathbf{r}'}\gamma(x;x')\right]_{x'=x}.$$

Do not replace the RDM by its diagonal too early

$[\nabla_{\mathbf{r}}^2\gamma(x;x')]_{x'=x}$ is not $\nabla_{\mathbf{r}}^2 n(x)$. The kinetic energy generally requires off-diagonal information; the local external-potential term does not.

Why the local density alone does not give the kinetic energy

Already for one particle, take a normalized orbital

$$\phi(\mathbf{r}) = \sqrt{n(\mathbf{r})}e^{i\theta(\mathbf{r})}.$$

Different phases θ give the *same* local density $|\phi|^2 = n$. But

$$\begin{aligned} T &= \frac{\hbar^2}{2m}\int \mathrm{d}^3 r\, |\nabla\phi|^2 \\ &= \frac{\hbar^2}{2m}\int \mathrm{d}^3 r\, \left[|\nabla\sqrt{n}|^2 + n|\nabla\theta|^2\right]. \end{aligned}$$

The one-body RDM retains the missing phase information:

$$\gamma(\mathbf{r};\mathbf{r}') = \sqrt{n(\mathbf{r})n(\mathbf{r}')}e^{i[\theta(\mathbf{r})-\theta(\mathbf{r}')]}.$$

Scope of the statement

This concerns arbitrary states. Ground-state density-functional theory introduces a separate constrained minimization; it does not replace $\gamma(x;x')$ by $n(x)$ inside the kinetic-energy formula.

First quantization III: derive the diagonal pair density

Count distinct particles in an *ordered* pair:

$$\widehat{P}(x,y) = \sum_{i \neq j} \delta(x - x_i)\delta(y - x_j).$$

As before, all $N(N-1)$ contributions have the same expectation:

$$\begin{aligned} P(x,y) &= \langle \widehat{P}(x,y) \rangle \\ &= N(N-1) \int \mathrm{d}x_3 \cdots \mathrm{d}x_N \, |\Psi(x,y,x_3,\ldots,x_N)|^2. \end{aligned}$$

$$\int \mathrm{d}x \, \mathrm{d}y \, P(x,y) = N(N-1), \qquad \int \mathrm{d}y \, P(x,y) = (N-1)n(x).$$

Number density versus probability density

$P/[N(N-1)]$ is the normalized joint density of two selected distinct particle labels. Antisymmetry implies $P(x,x) = 0$ for identical full space–spin coordinates.

The full two-body RDM and its contraction

Keep different coordinates on the bra and ket sides:

$$\boxed{\Gamma(x,y;x',y') = N(N-1)\int \mathrm{d}X\, \Psi(x,y,X)\Psi^*(x',y',X).}$$

Its diagonal is $P(x,y) = \Gamma(x,y;x,y)$, and

$$\Gamma(x,y;x',y')^* = \Gamma(x',y';x,y).$$

Tracing out one further particle gives

$$\begin{aligned}\int \mathrm{d}y\, \Gamma(x,y;x',y) &= N(N-1)\int \mathrm{d}y\, \mathrm{d}X\, \Psi(x,y,X)\Psi^*(x',y,X)\\ &= (N-1)\gamma(x;x').\end{aligned}$$

Consistency checks

$\mathrm{Tr}\,\gamma = N$, $\mathrm{Tr}\,\Gamma = N(N-1)$, and the contraction above must agree. These checks expose missing factors of two immediately.

The exact two-body energy from the first-quantized wave function

For symmetric local v, every unordered particle pair contributes equally:

$$\begin{aligned} E_2 &= \int \mathrm{d}x_1 \cdots \mathrm{d}x_N \, \Psi^* \sum_{i<j} v(\mathbf{r}_i, \mathbf{r}_j) \Psi \\ &= \frac{N(N-1)}{2} \int \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}X \, v(\mathbf{r}, \mathbf{r}') |\Psi(x, y, X)|^2 \\ &= \frac{1}{2} \int \mathrm{d}x \, \mathrm{d}y \, v(\mathbf{r}, \mathbf{r}') P(x, y). \end{aligned}$$

Exact energy of any normalized state

$$\boxed{E = T[\gamma] + \int \mathrm{d}^3 r \, v_{\mathrm{ext}}(\mathbf{r}) n(\mathbf{r}) + \frac{1}{2} \int \mathrm{d}x \, \mathrm{d}y \, v(\mathbf{r}, \mathbf{r}') P(x, y).}$$

No mean-field approximation has been made.

For a nonlocal or spin-changing interaction, its kernel must be contracted with the full off-diagonal Γ, not only with P.

Second quantization I: field operators and their action

In a complete orthonormal spin-orbital basis,

$$\widehat{\psi}(x) = \sum_p \phi_p(x) a_p, \qquad \widehat{\psi}^\dagger(x) = \sum_p \phi_p^*(x) a_p^\dagger,$$

$$\{\widehat{\psi}(x), \widehat{\psi}^\dagger(y)\} = \delta(x-y), \qquad \{\widehat{\psi}(x), \widehat{\psi}(y)\} = 0.$$

The N-particle state can be written as

$$|\Psi_N\rangle = \frac{1}{\sqrt{N!}} \int \mathrm{d}x_1 \cdots \mathrm{d}x_N \, \Psi(x_1, \ldots, x_N) \widehat{\psi}^\dagger(x_1) \cdots \widehat{\psi}^\dagger(x_N) |0\rangle.$$

Anticommuting $\widehat{\psi}(x)$ through the creators gives N equal terms:

$$\widehat{\psi}(x)|\Psi_N\rangle = \sqrt{N}\,|\Psi(x,\cdot)\rangle, \qquad \widehat{\psi}(y)\widehat{\psi}(x)|\Psi_N\rangle = \sqrt{N(N-1)}\,|\Psi(x,y,\cdot)\rangle.$$

The residual states on the right are generally *unnormalized*.

Second quantization II: the same one-body RDM

Take the overlap of the one-particle-removed states:

$$\begin{aligned}\langle\widehat{\psi}^\dagger(x')\widehat{\psi}(x)\rangle &= \left(\widehat{\psi}(x')|\Psi\rangle\right)^\dagger \widehat{\psi}(x)|\Psi\rangle \\ &= N\int \mathrm{d}X\, \Psi^*(x',X)\Psi(x,X) = \gamma(x;x').\end{aligned}$$

Thus

$$\boxed{\gamma(x;x') = \langle\widehat{\psi}^\dagger(x')\widehat{\psi}(x)\rangle, \qquad \widehat{n}(x) = \widehat{\psi}^\dagger(x)\widehat{\psi}(x).}$$

In a basis, define $D_{pq} = \langle a_q^\dagger a_p\rangle$; then

$$\gamma(x;x') = \sum_{pq}\phi_p(x)D_{pq}\phi_q^*(x'), \qquad \widehat{n}(x) = \sum_{pq}\phi_p^*(x)\phi_q(x)a_p^\dagger a_q.$$

The first- and second-quantized definitions are the same object, not two different approximations.

Second quantization III: pair density and normal ordering

The overlap of the two-particle-removed states gives

$$\boxed{\Gamma(x,y;x',y') = \langle \widehat{\psi}^\dagger(x')\widehat{\psi}^\dagger(y')\widehat{\psi}(y)\widehat{\psi}(x)\rangle.}$$

Using the field anticommutator,

$$\begin{aligned}\widehat{n}(x)\widehat{n}(y) &= \widehat{\psi}^\dagger(x)\widehat{\psi}(x)\widehat{\psi}^\dagger(y)\widehat{\psi}(y) \\ &= \delta(x-y)\widehat{\psi}^\dagger(x)\widehat{\psi}(y) + \widehat{\psi}^\dagger(x)\widehat{\psi}^\dagger(y)\widehat{\psi}(y)\widehat{\psi}(x).\end{aligned}$$

Consequently, as a distribution,

$$\boxed{P(x,y) = \langle \widehat{n}(x)\widehat{n}(y)\rangle - \delta(x-y)n(x).}$$

The delta term removes counting a particle paired with itself. In general $\langle \widehat{n}(x)\widehat{n}(y)\rangle \neq n(x)n(y)$.

The Hamiltonian in field operators

For the same local interaction as before,

$$\widehat{H}_1 = \int \mathrm{d}x\, \widehat{\psi}^\dagger(x)\widehat{h}_x\widehat{\psi}(x),$$
$$\widehat{H}_2 = \frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\, \widehat{\psi}^\dagger(x)\widehat{\psi}^\dagger(y)v(\mathbf{r},\mathbf{r}')\widehat{\psi}(y)\widehat{\psi}(x).$$

Taking expectation values immediately yields

$$E_1 = \int \mathrm{d}x\, [\widehat{h}_x\gamma(x;x')]_{x'=x}, \qquad E_2 = \frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\, v(\mathbf{r},\mathbf{r}')\Gamma(x,y;x,y).$$

What has been proved

The RDM energy expression follows both by integrating the many-body wave function and by taking field-operator expectation values. The factor $1/2$ counts each physical pair once.

The same exact energy in a discrete basis

Define *unsymmetrized* matrix elements

$$h_{pq} = \int \mathrm{d}x\, \phi_p^*(x)\widehat{h}_x\phi_q(x), \qquad v_{pq;rs} = \int \mathrm{d}x\, \mathrm{d}y\, \phi_p^*(x)\phi_q^*(y)v(\mathbf{r},\mathbf{r}')\phi_r(x)\phi_s(y).$$

Then

$$\widehat{H} = \sum_{pq} h_{pq}a_p^\dagger a_q + \frac{1}{2}\sum_{pqrs} v_{pq;rs}a_p^\dagger a_q^\dagger a_s a_r.$$

With $D_{pq} = \langle a_q^\dagger a_p\rangle$ and $\Gamma_{pq;rs} = \langle a_r^\dagger a_s^\dagger a_q a_p\rangle$,

$$\boxed{E = \sum_{pq} h_{pq}D_{qp} + \frac{1}{2}\sum_{pqrs} v_{pq;rs}\Gamma_{rs;pq}.}$$

Equivalently, $\bar{v}_{pq;rs} = v_{pq;rs} - v_{pq;sr}$ gives $\widehat{H}_2 = \frac{1}{4}\sum \bar{v}_{pq;rs}a_p^\dagger a_q^\dagger a_s a_r$. Do not use 1/2 and antisymmetrized matrix elements together here.

One determinant: derive its one-body RDM in first quantization

Let the occupied spin orbitals be orthonormal, and

$$\Phi(x_1, \ldots, x_N) = \frac{1}{\sqrt{N!}} \det[\phi_i(x_j)].$$

Expand the determinant along particle coordinate x:

$$\Phi(x, X) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} (-1)^{i+1} \phi_i(x) \Phi_{\widehat{i}}(X).$$

$\Phi_{\widehat{i}}$ is the normalized determinant with orbital i omitted; orthonormality implies $\int \mathrm{d}X \, \Phi^*_{\widehat{j}} \Phi_{\widehat{i}} = \delta_{ij}$. Substitution gives

$$\begin{aligned} \gamma_\Phi(x; x') &= \sum_{ij} (-1)^{i+j} \phi_i(x) \phi_j^*(x') \delta_{ij} \\ &= \boxed{\sum_{i \in \mathrm{occ}} \phi_i(x) \phi_i^*(x')}. \end{aligned}$$

Therefore $n(x) = \sum_{i \in \mathrm{occ}} |\phi_i(x)|^2$ and $\gamma_\Phi^2 = \gamma_\Phi$.

One determinant: derive its two-body RDM in first quantization

Expand along two particle coordinates, with $X = (x_3, \ldots, x_N)$:

$$\Phi(x, y, X) = \frac{1}{\sqrt{N(N-1)}} \sum_{i<j} s_{ij} A_{ij}(x, y) \Phi_{\widetilde{ij}}(X),$$

$$A_{ij}(x, y) = \phi_i(x)\phi_j(y) - \phi_j(x)\phi_i(y), \qquad s_{ij} = \pm 1.$$

The normalized remaining determinants are orthogonal. Their integration therefore cancels the prefactor in the definition of Γ:

$$\begin{aligned}
\Gamma_\Phi(x, y; x', y') &= \sum_{i<j} A_{ij}(x, y) A^*_{ij}(x', y') \\
&= \sum_{ij} \phi_i(x)\phi_j(y)[\phi^*_i(x')\phi^*_j(y') - \phi^*_j(x')\phi^*_i(y')] \\
&= \boxed{\gamma_\Phi(x; x')\gamma_\Phi(y; y') - \gamma_\Phi(x; y')\gamma_\Phi(y; x')}.
\end{aligned}$$

The $i = j$ terms cancel. The minus sign is a consequence of antisymmetry.

The same direct-minus-exchange result in second quantization

Write $|\Phi\rangle = \prod_{i\in\mathrm{occ}} a_i^\dagger |0\rangle$ in its occupied basis. The occupations are $n_i = 0$ or 1, and

$$\langle a_i^\dagger a_k \rangle = \delta_{ik} n_i.$$

For $i \neq j$, $a_l a_k$ must remove the same pair that $a_i^\dagger a_j^\dagger$ restores. There are only two possibilities:

$$\langle a_i^\dagger a_j^\dagger a_l a_k \rangle = n_i n_j (\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}).$$

The exchanged removal order gives one extra fermionic minus sign; for $i = j$ the whole expression vanishes. Equivalently (Wick's theorem),

$$\langle a_r^\dagger a_s^\dagger a_q a_p \rangle = \langle a_r^\dagger a_p \rangle \langle a_s^\dagger a_q \rangle - \langle a_r^\dagger a_q \rangle \langle a_s^\dagger a_p \rangle.$$

Inserting the field expansions gives exactly the factorization derived from determinant minors on the previous slide.

The determinant pair density and its normalization

Set $x' = x$, $y' = y$ and use Hermiticity:

$$\boxed{P_\Phi(x,y) = n(x)n(y) - |\gamma_\Phi(x;y)|^2.}$$

- The direct term uses the local one-body densities.
- The exchange term uses the off-diagonal one-body density matrix.
- At identical full coordinates, $P_\Phi(x,x) = n(x)^2 - n(x)^2 = 0$.

The trace checks the coefficient:

$$\int \mathrm{d}x\,\mathrm{d}y\, P_\Phi(x,y) = N^2 - \mathrm{Tr}\,\gamma_\Phi^2 = N^2 - N = N(N-1).$$

The assumption is essential

This factorization is exact for a Slater determinant. Substituting a correlated state's γ into it generally does not give its exact Γ.

For a pure fixed-N state, $\gamma^2 = \gamma$ exactly when it is a single determinant.

Insert the pair density: Hartree and exchange energies

For one determinant, the interaction energy becomes

$$E_2[\Phi] = \frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\; v(\mathbf{r},\mathbf{r}')n(x)n(y) - \frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\; v(\mathbf{r},\mathbf{r}')|\gamma_\Phi(x;y)|^2 \equiv E_H[n] + E_x[\gamma_\Phi].$$

Because v does not act on spin,

$$\boxed{E_H[n] = \frac{1}{2}\int \mathrm{d}^3r\,\mathrm{d}^3r'\; n(\mathbf{r})v(\mathbf{r},\mathbf{r}')n(\mathbf{r}').}$$

Thus

$$\boxed{E[\Phi] = T[\gamma_\Phi] + E_{\mathrm{ext}}[n] + E_H[n] + E_x[\gamma_\Phi].}$$

This is the exact expectation value in the chosen determinant. **Hartree–Fock** further minimizes it over orthonormal occupied orbitals.

Spin resolution: when does exchange act?

The general spinor expression is

$$E_x = -\frac{1}{2}\sum_{\sigma\sigma'}\int \mathrm{d}^3r\,\mathrm{d}^3r'\; v(\mathbf{r},\mathbf{r}')|\gamma(\mathbf{r}\sigma;\mathbf{r}'\sigma')|^2.$$

For a collinear determinant built from definite-spin orbitals,

$$\gamma(\mathbf{r}\sigma;\mathbf{r}'\sigma') = \delta_{\sigma\sigma'}\gamma_\sigma(\mathbf{r};\mathbf{r}'), \quad E_x = -\frac{1}{2}\sum_{\sigma}\int \mathrm{d}^3r\,\mathrm{d}^3r'\; v|\gamma_\sigma|^2.$$

The corresponding spin-summed spatial pair density is

$$P_{\mathrm{spatial}}(\mathbf{r},\mathbf{r}') = n(\mathbf{r})n(\mathbf{r}') - \sum_{\sigma}|\gamma_\sigma(\mathbf{r};\mathbf{r}')|^2.$$

Pauli exclusion concerns full space–spin coordinates

Equal-spin coincidence vanishes. Opposite-spin particles may have nonzero pair density at the same spatial position. For general noncollinear spinors, use the full spin-matrix expression above.

The Hartree term produces a local mean field

Define

$$v_H(\mathbf{r}) = \int \mathrm{d}^3 r'\, v(\mathbf{r}, \mathbf{r}') n(\mathbf{r}'), \qquad E_H = \frac{1}{2} \int \mathrm{d}^3 r\, n(\mathbf{r}) v_H(\mathbf{r}).$$

Varying both density factors gives, by symmetry of v,

$$\begin{aligned} \delta E_H &= \frac{1}{2} \int \mathrm{d}^3 r\, \mathrm{d}^3 r'\, v[\delta n(\mathbf{r}) n(\mathbf{r}') + n(\mathbf{r}) \delta n(\mathbf{r}')] \\ &= \int \mathrm{d}^3 r\, v_H(\mathbf{r}) \delta n(\mathbf{r}), \qquad \boxed{\frac{\delta E_H}{\delta n(\mathbf{r})} = v_H(\mathbf{r}).} \end{aligned}$$

For electrons, $v(\mathbf{r}, \mathbf{r}') = e^2/(4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|)$.

Why local?

The Hartree operator multiplies an orbital by $v_H(\mathbf{r})$ at that same position. Its value depends on the density everywhere, but its action on an orbital is local.

The exchange term produces a nonlocal Fock operator

Vary an occupied orbital $\phi_i^*(x)$, keeping the orbital normalization constraints. The two equal exchange contributions cancel the factor $1/2$:

$$\frac{\delta E_x}{\delta \phi_i^*(x)} = -\int \mathrm{d}y\, v(\mathbf{r},\mathbf{r}')\gamma(x;y)\phi_i(y).$$

Combining all variations gives

$$\underbrace{\widehat{h}_x\phi_i(x) + v_H(\mathbf{r})\phi_i(x) - \int \mathrm{d}y\, v(\mathbf{r},\mathbf{r}')\gamma(x;y)\phi_i(y)}_{(\widehat{f}\phi_i)(x)} = \sum_j \epsilon_{ij}\phi_j(x).$$

Diagonalizing the Hermitian constraint matrix yields canonical HF orbitals, $\widehat{f}\phi_i = \epsilon_i\phi_i$.

Why nonlocal?

Exchange samples $\phi_i(y)$ at other coordinates y. In general it cannot be replaced by multiplying by a simple function of the local density $n(\mathbf{r})$.

Recover the familiar direct and exchange matrix elements

With occupied spin orbitals, define

$$J_{ij} = \int \mathrm{d}x\,\mathrm{d}y\,|\phi_i(x)|^2 v(\mathbf{r},\mathbf{r}')|\phi_j(y)|^2,$$
$$K_{ij} = \int \mathrm{d}x\,\mathrm{d}y\,\phi_i^*(x)\phi_j^*(y)v(\mathbf{r},\mathbf{r}')\phi_j(x)\phi_i(y).$$

Substituting the orbital expansion of γ gives

$$E_H = \frac{1}{2}\sum_{ij\in\mathrm{occ}} J_{ij}, \qquad E_x = -\frac{1}{2}\sum_{ij\in\mathrm{occ}} K_{ij},$$

$$\boxed{E[\Phi] = \sum_{i\in\mathrm{occ}} h_{ii} + \frac{1}{2}\sum_{ij\in\mathrm{occ}} (J_{ij} - K_{ij}).}$$

Explicitly, $J_{ij} - K_{ij} = v_{ij;ij} - v_{ij;ji} = \bar{v}_{ij;ij}$. The coordinate-space density derivation is the same direct-minus-exchange result used in the earlier second-quantization lectures.

A useful check: exchange removes one-particle self-interaction

For a single occupied spin orbital ϕ,

$$n(x) = |\phi(x)|^2, \qquad \gamma(x;y) = \phi(x)\phi^*(y).$$

Then

$$P(x,y) = |\phi(x)|^2|\phi(y)|^2 - |\phi(x)\phi^*(y)|^2 = 0, \qquad E_H + E_x = 0.$$

More generally, $J_{ii} = K_{ii}$: each diagonal self term cancels.

The direct term alone is not the exact fixed-N pair density

The product $n(x)n(y)$ counts self pairs. Even a distinguishable-particle Hartree product has $\sum_{i\neq j} |\phi_i(x)|^2|\phi_j(y)|^2$, not generally $n(x)n(y)$. For fermions, determinant exchange supplies the required self-interaction cancellation and additional antisymmetry effects.

Worked example: two opposite spins in one spatial orbital

For normalized $\varphi(\mathbf{r})$, occupy $\varphi\alpha$ and $\varphi\beta$:

$$\Phi(x_1, x_2) = \frac{\varphi(\mathbf{r}_1)\varphi(\mathbf{r}_2)}{\sqrt{2}}[\alpha(\sigma_1)\beta(\sigma_2) - \beta(\sigma_1)\alpha(\sigma_2)].$$

$$n_\uparrow = n_\downarrow = |\varphi|^2, \quad n = 2|\varphi|^2, \qquad \gamma(\mathbf{r}\uparrow; \mathbf{r}'\downarrow) = 0.$$

For our ordered-pair convention,

$$\boxed{P(\mathbf{r}\uparrow, \mathbf{r}'\downarrow) = |\varphi(\mathbf{r})|^2 |\varphi(\mathbf{r}')|^2.}$$

Let $J = \int \mathrm{d}^3r\, \mathrm{d}^3r' |\varphi(\mathbf{r})|^2 v |\varphi(\mathbf{r}')|^2$. Then

$$E_H = 2J, \qquad E_x = -J, \qquad E_2 = J.$$

There is no opposite-spin exchange contraction, yet exchange cancels the two same-spin self terms. Spatial coincidence density can be nonzero.

Worked example: parallel spins and the meaning of coincidence

For two orthonormal spatial orbitals with both spins up,

$$\Phi(\mathbf{r}_1\uparrow,\mathbf{r}_2\uparrow)=\frac{\varphi_1(\mathbf{r}_1)\varphi_2(\mathbf{r}_2)-\varphi_2(\mathbf{r}_1)\varphi_1(\mathbf{r}_2)}{\sqrt{2}}.$$

Since $N(N-1)=2$,

$$P(\mathbf{r}_1\uparrow,\mathbf{r}_2\uparrow)=|\varphi_1(\mathbf{r}_1)\varphi_2(\mathbf{r}_2)-\varphi_2(\mathbf{r}_1)\varphi_1(\mathbf{r}_2)|^2,\quad P(\mathbf{r}\uparrow,\mathbf{r}\uparrow)=0.$$

A continuous density is not a probability at an exact point

The probability for two selected distinct labels to lie in finite space–spin cells A, B is

$$\Pr(x_1\in A,x_2\in B)=\frac{1}{N(N-1)}\int_A \mathrm{d}x\int_B \mathrm{d}y\,P(x,y).$$

An exact point has zero measure. Opposite-spin coincidence means a potentially nonzero *density*, not unit probability.

Beyond one determinant: the two-body cumulant

For an arbitrary state, define the residual two-body information by

$$\Gamma(x,y;x',y') = \gamma(x;x')\gamma(y;y') - \gamma(x;y')\gamma(y;x') + \lambda(x,y;x',y').$$

This is an exact decomposition, with $\lambda = 0$ for a determinant. Its contraction is

$$\int \mathrm{d}y\, \lambda(x,y;x',y) = (\gamma^2 - \gamma)(x;x'),$$

not generally zero for a correlated state. The exact energy can be organized as

$$\boxed{E = T[\gamma] + E_{\mathrm{ext}}[n] + E_H[n] + E_x[\gamma] + E_\lambda,} \qquad E_\lambda = \frac{1}{2}\int \mathrm{d}x\,\mathrm{d}y\, v\lambda(x,y;x,y).$$

What this correlation contribution means

E_λ is the interaction-cumulant contribution for the exact γ. It is not by itself $E_{\mathrm{exact}} - E_{\mathrm{HF}}$: the exact and HF one-body RDMs, and hence their kinetic and other energy contributions, can differ too.

Why the exact two-body RDM is not fixed by a generic one-body RDM

Take four orthonormal spin orbitals and two fermions:

$$|12\rangle = a_1^\dagger a_2^\dagger |0\rangle, \quad |34\rangle = a_3^\dagger a_4^\dagger |0\rangle, \quad |\Psi_\pm\rangle = \frac{|12\rangle \pm |34\rangle}{\sqrt{2}}.$$

A one-body operator cannot connect determinants differing in two occupations. Both states therefore have

$$D^{(+)}_{pq} = D^{(-)}_{pq} = \frac{1}{2}\delta_{pq}, \qquad p, q = 1, \ldots, 4.$$

But a pair-transfer operator does distinguish them:

$$\langle a_3^\dagger a_4^\dagger a_2 a_1 \rangle_{\Psi_+} = \frac{1}{2}, \qquad \langle a_3^\dagger a_4^\dagger a_2 a_1 \rangle_{\Psi_-} = -\frac{1}{2}.$$

Thus identical one-body RDMs (and local densities) can accompany different two-body RDMs and different energies for an interaction coupling these pairs.

The exchange hole: one particle is excluded from the neighborhood

For $n(x) > 0$, define the conditional density of other particles as

$$n_{\mathrm{cond}}(y|x) = \frac{P(x,y)}{n(x)}, \qquad n_{xc}(x,y) = \frac{P(x,y)}{n(x)} - n(y).$$

The exact contraction gives

$$\int \mathrm{d}y\, n_{xc}(x,y) = (N-1) - N = -1.$$

For a determinant this hole is entirely exchange:

$$n_x(x,y) = -\frac{|\gamma_\Phi(x;y)|^2}{n(x)}, \qquad \int \mathrm{d}y\, n_x = -\frac{(\gamma_\Phi^2)(x;x)}{n(x)} = -1.$$

For correlated states, exchange-like and cumulant pieces together enforce the exact sum rule. The extra opposite-spin correlation hole is absent from the closed-shell determinant example but can appear in correlated methods.

How this connects to density-functional theory

For arbitrary states, the explicit exact expression is $E[\gamma, \Gamma]$; the local density by itself does not specify those RDMs.

Ground-state constrained search (fixed interaction and N)

$$F[n] = \min_{\Psi \to n} \langle \widehat{T} + \widehat{V} \rangle_\Psi, \qquad E_0 = \min_n \left\{ F[n] + \int \mathrm{d}^3 r \, v_{\mathrm{ext}}(\mathbf{r}) n(\mathbf{r}) \right\}.$$

The minimizing wave function supplies the missing information implicitly.

The Kohn–Sham organization is

$$E[n] = T_s[n] + E_{\mathrm{ext}}[n] + E_H[n] + E_{xc}[n].$$

E_{xc} contains the interaction beyond Hartree and the difference $T - T_s$. It is not just the two-body cumulant energy E_λ.

This reconciles the density-matrix derivation with the subsequent Hohenberg–Kohn and Kohn–Sham lectures; it does not provide an explicit formula for the unknown exact functional.

A compact checklist for calculations

1. **Fix conventions:** $x = (\mathbf{r}, \sigma)$; ket coordinates first; $\mathrm{Tr}\,\gamma = N$, $\mathrm{Tr}\,\Gamma = N(N-1)$.
2. **Any state:**
$$\gamma = \langle \widehat{\psi}^\dagger \widehat{\psi} \rangle, \quad \Gamma = \langle \widehat{\psi}^\dagger \widehat{\psi}^\dagger \widehat{\psi} \widehat{\psi} \rangle, \qquad E = \mathrm{Tr}(h\gamma) + \tfrac{1}{2}\,\mathrm{Tr}(v\Gamma).$$
3. **One determinant:**
$$\Gamma = \gamma\gamma - \gamma\gamma\big|_{\mathrm{exchange}}, \qquad E = T + E_{\mathrm{ext}} + E_H + E_x.$$
4. **Beyond a determinant:** retain the additional two-body information λ; do not assume Wick factorization.
5. **Check limiting cases:** one-particle $E_H + E_x = 0$; same-spin coincidence vanishes; opposite-spin coincidence need not.

Answer to the original question

The Hartree term is an explicit functional of the local density. Kinetic and exchange terms require the one-body RDM; the exact correlated interaction additionally requires the two-body RDM.

Sources and further reading

- M. Hjorth-Jensen, *FYS4480/9480, Week 42*, supplied course slides, especially slides 31–34 (and the follow-up answer on slide 35). The pair-normalization conventions are reconciled explicitly in this supplement.
- Nuclear TALENT, *Reminder on Second Quantization*, operator conventions and expectation values. nucleartalent.github.io/NuclearStructure: Second Quantization.
- Nuclear TALENT, *Hartree–Fock*, direct/exchange energy and variation. nucleartalent.github.io/NuclearStructure: Hartree–Fock.
- A. L. Fetter and J. D. Walecka, *Quantum Theory of Many-Particle Systems* (McGraw–Hill, 1971; Dover reprint).